时茗园智慧茶园 BI 系统 - 数据结构与接口规范

1. 核心数据对象模型

1.1 指标快照对象 (IndicatorSnapshot)

用于大屏实时展示的数据模型。

```
interface IndicatorSnapshot {
  timestamp: number;
  ecoHealth: {
    totalScore: number;
    biodiversity: number;
    soilHealth: number;
    airWaterQuality: number;
  };
  carbonSequestration: {
    currentTotal: number; // 当前固碳量 (吨)
    accumulatedValue: number; // 累计碳汇价值 (CNY)
  };
  growthMetrics: {
    photosynthesisIndex: number;
    growthMatchRate: number; // 生长节律匹配度
    microClimateBalance: number;
  };
  infrastructure: {
    iotDensity: number; // 物联感知密度 (个/公顷)
    digitalTwinAccuracy: number;
    responseTime: number; // AI 响应速度 (ms)
  };
}
```

1.2 预警事件对象 (WarningEvent)

```
interface WarningEvent {
  id: string;
  type: 'CLIMATE' | 'PEST' | 'IRRIGATION' | 'SOIL';
  level: 'INFO' | 'WARNING' | 'CRITICAL';
  message: string;
  leadTime: number; // 提前量 (秒)
  location: { x: number, y: number, z: number }; // 3D 空间坐标
  suggestedAction: string;
}
```

2. API 接口定义

2.1 获取实时大屏看板数据

- **Endpoint**: `GET /api/bi/cockpit/realtime`
- **Response**: `IndicatorSnapshot`

2.2 获取 3D 场景配置与状态

- **Endpoint**: `GET /api/bi/scene/config`
- **Response**: 包含地形模型 URL、材质配置、当前天气模拟参数。

2.3 预警列表流 (WebSocket)

- **Topic**: `/topic/bi/warnings`
- **Payload**: `WarningEvent[]`

3. 数据库表设计预览 (MySQL/PostgreSQL)

3.1 `bi_indicator_history` (历史指标表)

- `id` : BIGINT (PK)
- `category` : VARCHAR (生态/生长/中枢/价值)
- `indicator_key` : VARCHAR (e.g., 'carbon_sequestration')
- `indicator_value` : DOUBLE
- `record_time` : TIMESTAMP

3.2 `bi_warning_log` (预警日志表)

- `id` : BIGINT (PK)
- `event_type` : VARCHAR
- `warning_content` : TEXT
- `actual_occurrence_time` : TIMESTAMP
- `warning_trigger_time` : TIMESTAMP
- `status` : INT (已处理/未处理)